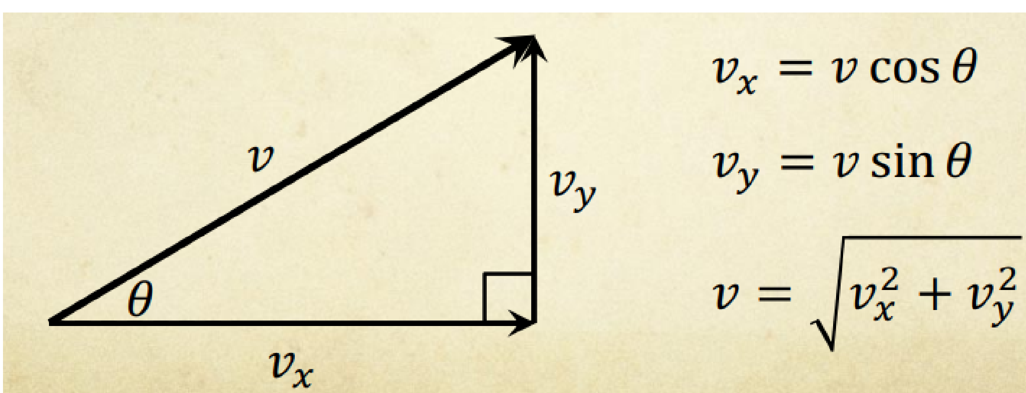
Projectile Motion

Projectiles

- A projectile is an object dropped or launched into a uniform gravitational field.
- The only force acting on the object is gravity.
    - Air resistance and friction is ignored.
- A projectile follows a curved path, more identifiably as a parabola.

Galileo's Observations and Galilean Motion

- Galileo performed experiments to establish laws of falling objects using an inclined plane.
- Galileo observed that all objects fall at the same rate and with constant acceleration.
    - He used this to establish the time-squared law, which states that $s \propto t^2$.
- Galileo's analysis of projectile motion showed that motion can be separated into independent horizontal and vertical components.
    - The horizontal component experiences constant velocity, while the vertical component experiences constant downwards acceleration.
- Velocity is a vector and can have any orientation, and can be resolved into horizontal and vertical components.



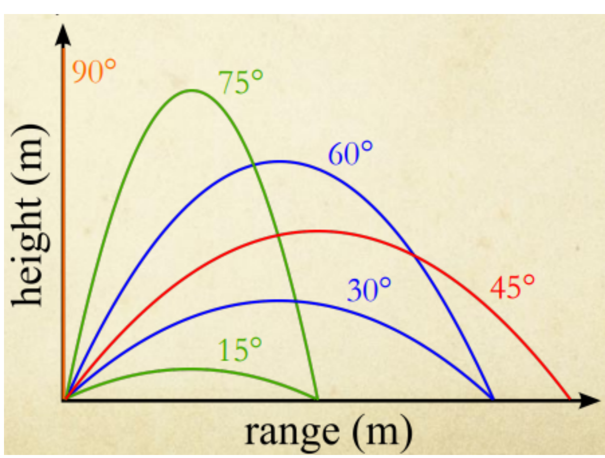
- For the horizontal component, no acceleration means that $u_x = v_x$.
  - v_x can also be calculated through $v_x = \frac{s}{t}$.
- The vertical component experiences uniform linear acceleration, so SUVAT equations involving acceleration can calculate unknown variables.
  - These equations are $v_y = u_y + a_y t$, $s_y = u_y t + \frac{1}{2} a_y t^2$ and $v^2_y = u^2_y + 2a_y s_y$.

Projectile Trajectory

- The net motion of a projectile is the combination of the horizontal and vertical components.
- The uniform horizontal motion and accelerating vertical motion results in a parabolic trajectory.
- Horizontal and vertical axes can be used as a reference, where the launch position is the origin.
  - The up and right directions are defined as positive.

Effect of Variables

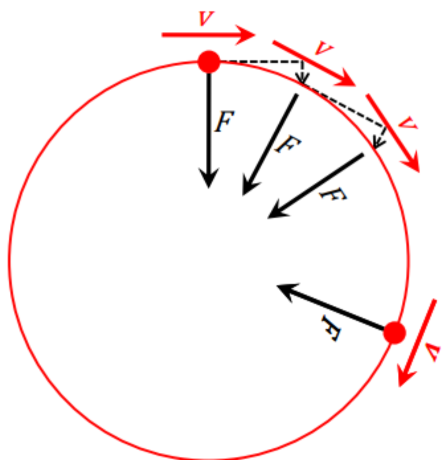
- The time of flight and height are only dependent on the u_y, where u_x has no effect, and the range of the projectile is determined by u_x and t.
- u_x and u_y are dependent on u and θ_u.
- For any given u, the maximum range occurs when $\theta = 45°$ and the maximum height occurs when $\theta = 90°$.



Circular Motion

The Acceleration Vector

- Acceleration is a vector according to $F_{\text{net}} = ma$.
  - If force is applied at a right angle to the direction the object is travelling, then circular motion occurs.

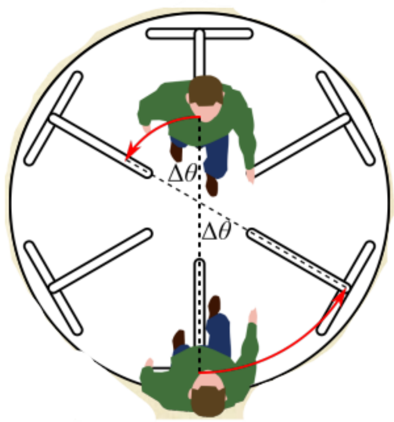


Uniform Circular Motion

- Uniform circular motion results from a constant net force perpendicular to the velocity at all times.
  - This force is called the centripetal force, and is denoted by the symbol F_c.
  - The force applied leads to a constant acceleration towards the centre of the circle called centripetal acceleration, and is denoted by the symbol a_c.
- Constant acceleration means there is a constant change in velocity.
  - The magnitude of velocity is constant, but the direction is constantly changing.
- Centripetal velocity is given by $a_c = \frac{v^2}{r}$.
  - Centripetal force is given by $F_c = ma_c = \frac{mv^2}{r}$, from Newton's 2nd Law.

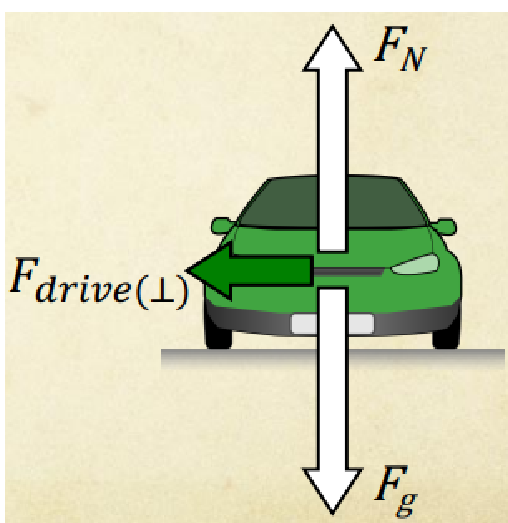
Angular Velocity

- Angular velocity is a measure of the rate of rotation, where $\omega = \frac{\Delta\theta}{t}$.
    - The units of angular velocity are commonly rads^{-1}, degs^{-1} or rpm.
- Velocity defined through angular velocity as $v = \omega r$.
    - If the rotation is clockwise, then the velocity is into the page, and vice versa.

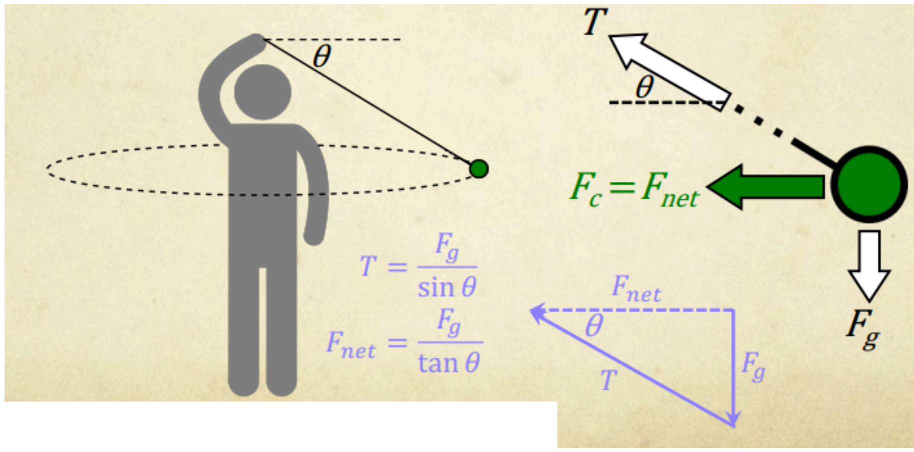


Applications of Circular Motion

- For a car driving at a constant speed around a curve in the road, the centripetal force is provided by friction.

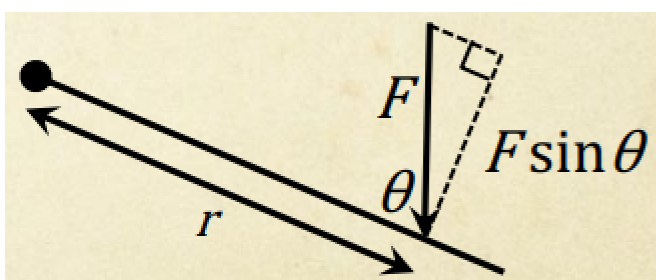


- For a mass being spun on the end of a string, the centripetal force is provided by the tension of the string.



Torque

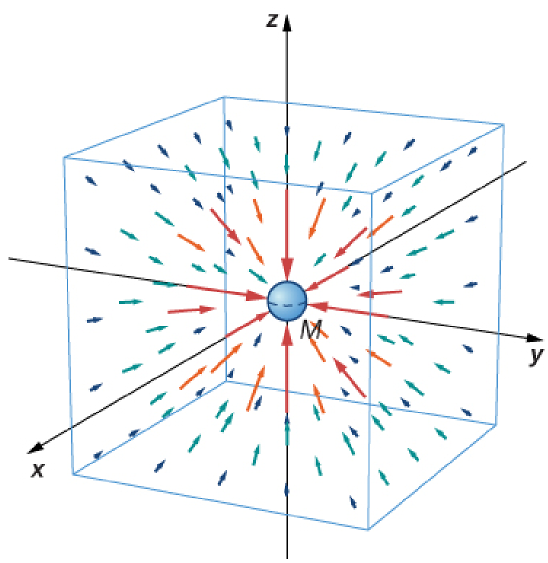
- The force that causes rotation is called torque, denoted by the symbol τ.
    - The units for torque are Nm, or Newton metres.
- Torque involves applying a force at a distance from a pivot point, where $\tau = r_{\perp}F$.
- The force must be perpendicular to the lever arm.
    - The lever arm is the line joining the point of application and the pivot.



Motion in Gravitational Fields

Gravitational Fields

- Gravity is the force of attraction between any two masses, and is associated with a gravitational field.
    - Gravity determines weight on Earth.
    - All masses are surrounded by a gravitational field.
- Gravitational fields are represented by field lines, where the direction is towards the centre of mass.
    - Field lines do not cross, and the closer the lines, the stronger the field is in that region.
    - Field lines represent the direction of the force applied on a mass.



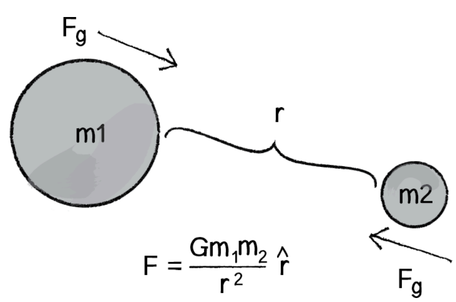
- On a human scale, the gravitational field of the Earth is observed as uniform.
    - This means that the gravitational field lines are parallel.
- On a planetary scale, Earth can be considered a sphere.
    - The gravitational field lines are directed radially inwards, and field strength decreases with distance from the Earth.

Universal Law of Gravitation

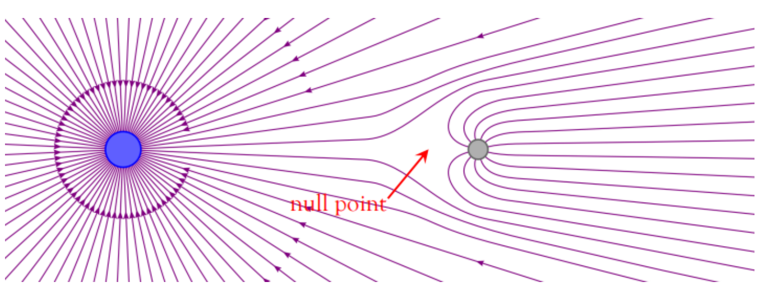
- Isaac Newton proposed his Law of Universal Gravitation in 1687, which describes the force between any two masses.
- There were three main features of the law:
  - Gravity is universal, meaning that every particle gravitationally attracts every other particle.
  - Gravitational force is directly proportional to the masses of the particles.
  - Gravitational force is inversely proportional to the square of the distance between their masses.
- The Universal Law of Gravitation can be described quantitatively as $F_g = \frac{GMm}{r^2}$.
  - G is the universal gravitational constant, equal to $6.67 \times 10^{-11}\ Nm^2kg^{-2}$.

Gravitational Force

- Gravitational force is the force of attraction between two masses, where the forces are equal and directed towards each other.
  - Both masses will move towards each other.



- Gravitation fields superimpose, where the field at any point is the vector sum of individual fields.



Gravitational Potential Energy

- Gravitational potential energy is defined as the work done to move an object from a very large distance away to a point in a gravitational field.
  - Near the Earth's surface, $\Delta U = mg\Delta h$, but further out curvature of the planet becomes a factor.
- Gravitational potential energy is quantitatively defined as $U =- \frac{GMm}{r}$.
  - U is negative because $U = 0$ when $r = \infty$, and K increases as the object falls, meaning that U decreases from the Law of Conservation of Energy.

Escape Velocity

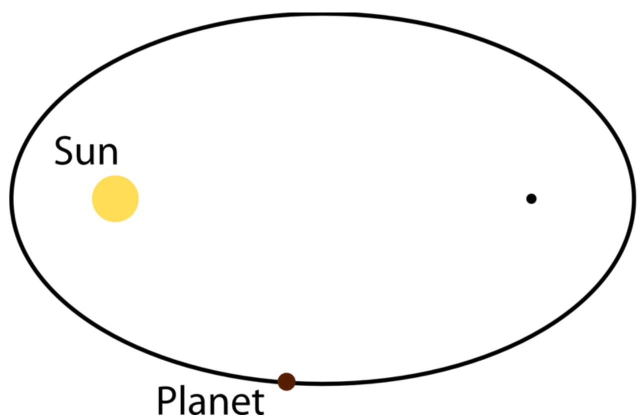
- The escape velocity is the velocity at which an object can be launched from a planet and escape its gravitational field.
- Escape velocity is defined by the relationship K of launch $= - U$ at the altitude of launch, where U is negative because K increases as U increases.
  - This relationship is quantitatively defined as $\frac{1}{2}mv^2 = \frac{GMm}{r}$, which leads to the equation $v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$.

Planetary Orbit

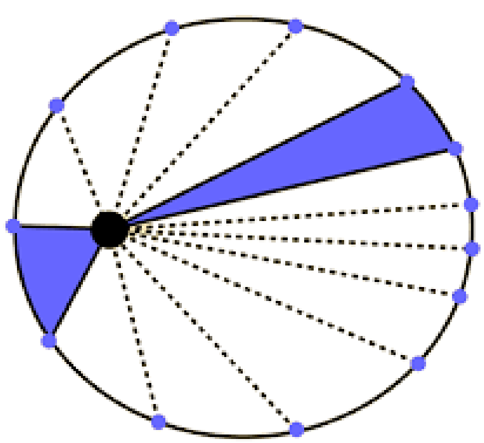
- In the circular orbit around a planet, the centripetal force is provided by the planet's gravity, meaning that $\frac{mv^2}{r} = \frac{GMm}{r^2}$.
  - This leads to the equation $v_{\text{o}} = \sqrt{\frac{GM}{r}}$, where v_{o} is the orbital velocity.
- For a circular orbit, the magnitude of the orbital velocity is constant.
  - This is shown in the equation $v_{\text{o}} = \frac{s}{t} = \frac{2\pi r}{T}$, where T is the orbital period.
  - The orbital period is the time taken to complete one orbit.
- The kinetic energy of an object in orbit is defined as $K = \frac{GMm}{2r}$.
  - This means that the total energy of an object in orbit is $E = - \frac{GMm}{2r}$.
- Because the centripetal force is always perpendicular to velocity, the planet's gravity is doing no work on the spacecraft.

Kepler's Laws of Planetary Motion

- Johannes Kepler published the three laws of planetary motion in 1609.
    - Kepler's First Law is the Law of Orbits.
    - Kepler's Second Law is the Law of Areas.
    - Kepler's Third Law is the Law of Periods.
- Kepler's Law of Orbits states that all planets move in elliptical orbits, with the sun at one of the two foci.



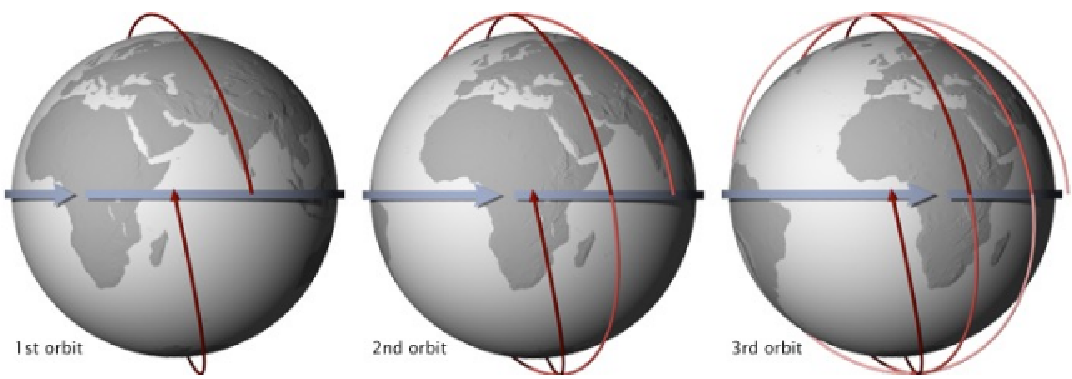
- Kepler's Law of Areas states that a line that connects a planet to the sun sweeps out equal areas in equal time intervals.
    - The orbital radius and angular velocity vary to conserve angular momentum.



- Kepler's Law of Periods states that the square of a planet's orbital period is directly proportional to the cube of the average radius of a planet's orbit.
  - This means that $T^2 \propto r^3$, or $T^2 = kr^3$ for some constant k.
- This relationship is further developed as $\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$.
  - While SI units are used, the other units for this law are astronomical units, years and solar masses.

Low Earth and Geostationary Orbits

- Low Earth orbits have an altitude limited between 200 and 1000 km.
  - This range is to prevent atmospheric drag from a low altitude, and to avoid van Allen radiation belts at a high altitude.
- Low Earth orbits allow the viewing of Earth's surface after several orbits.



- Low Earth orbits lie within the upper limits of the Earth's atmosphere, or the mesosphere, so it experiences a small amount of atmospheric drag.
  - This decrease in velocity means that the satellite will lose altitude, eventually experiencing orbital decay.

- Geosynchronous orbits' orbital period matches the rotation of the Earth.
  - Their orbital period is approximately 1436 minutes or 23.93 hours, but 24 hours can be used, which is called a sidereal day.

    - The period is not exactly 24 hours due to the orbital motion of the Earth around the sun.
- A geostationary orbit is a specific geosynchronous orbit situated above the equator.
    - It appears stationary in the sky when viewed from the surface of the Earth.
    - This orbit is also referred to as a geosynchronous equatorial orbit.
- A summary table comparing information between low Earth orbits and geostationary orbits is below.

	Low Earth Orbit	**Geostationary Orbit**
Altitude (km)	200 - 1 000	35 786
Orbital Period (min)	90 - 105	1436
Orbital Velocity (kms^{-1})	7.8 - 7.4	3.1
Motion Relative to the Surface of the Earth	Moving	Stationary
View of the Earth's Surface	Limited, but the entire surface can be viewed over several orbits	Large, but limited to a third of the Earth's surface
Uses	• High resolution imaging • Manned spacecraft	• Telecommunications • Weather tracking • Not GPS
Advantages	• Higher detail view of the Earth • Less fuel required • Faster mapping of the Earth	• Easy to locate • Does not experience orbital decay
Disadvantages	• More difficult to track	• Signal delay must be accounted for

	• Atmospheric drag results in orbital decay • More easily affected by changes in van Allen belts • Interferes with other satellites	• Limited view, so more satellites needed to map the Earth • More expensive • Exposed to higher radiation levels.